The Pol-InSAR Course

Part 5: SAR Interferometry 3

Prepared by DLR-HR's Pol-InSAR Team

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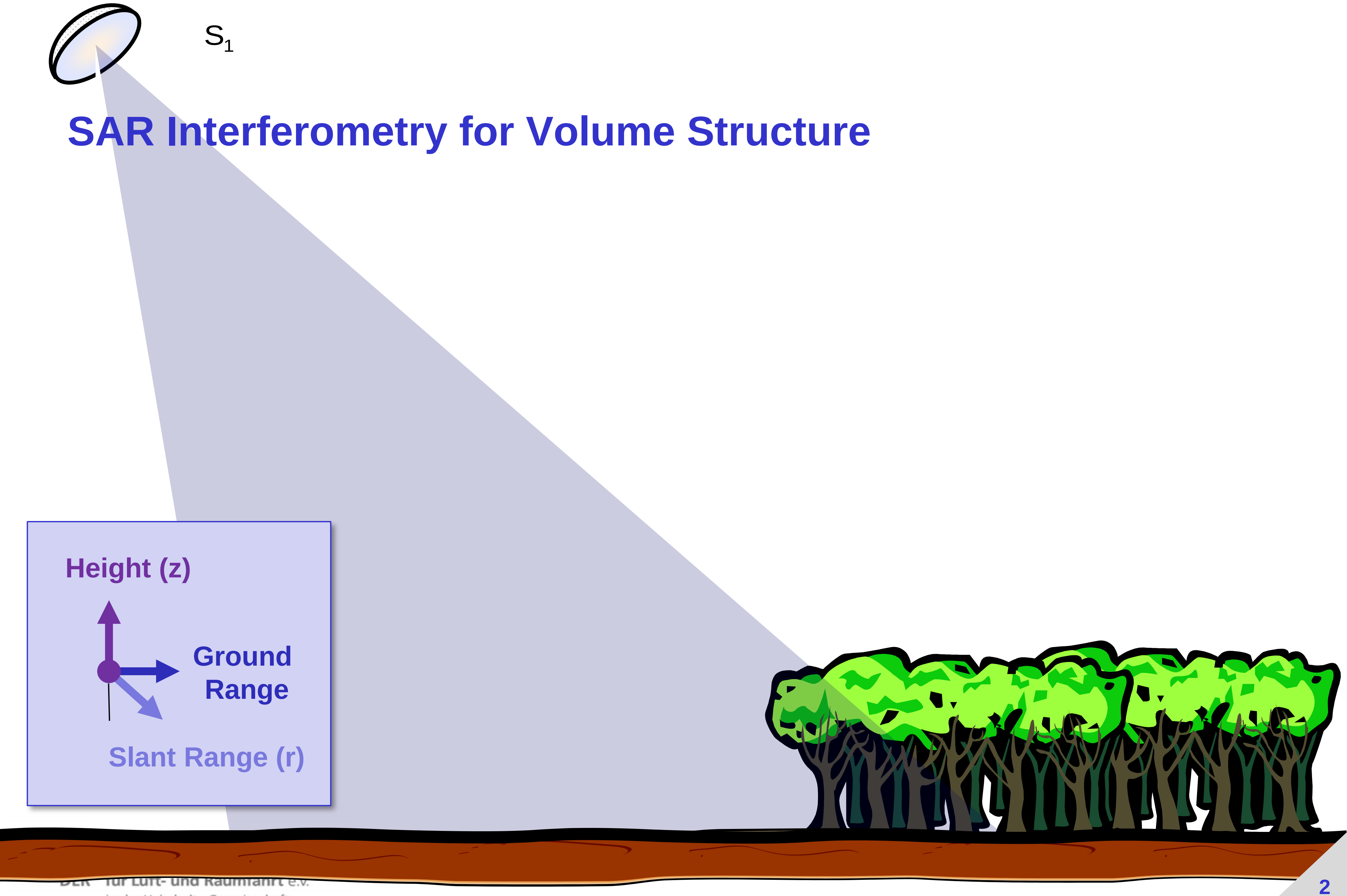
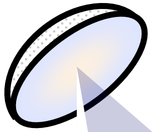
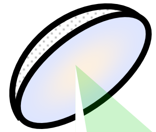
DLR esa ESAMAAP EEBIOMASS

SAR Interferometry for Volume Structure





S_1 S_2



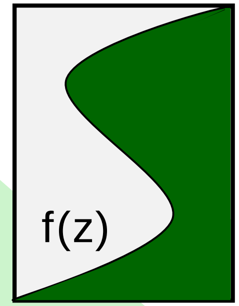
Interferometric Coherence

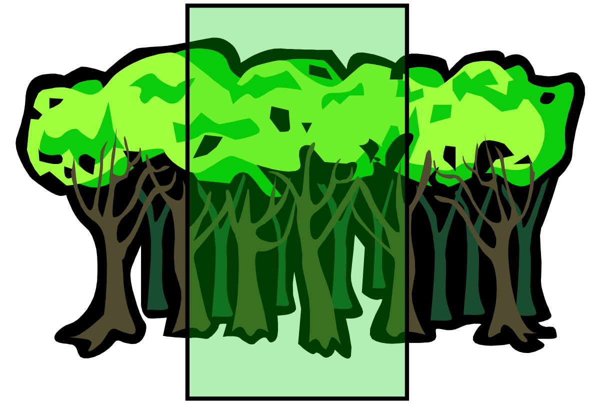
$$\tilde{\gamma}(S_1\ S_2) = \frac{< S_1\ S_2^* >}{\sqrt{< S_1\ S_1^* > < S_2\ S_2^* >}}$$

SAR Interferometry for Volume Structure

Volume Coherence

$$\tilde{\gamma}_{Vol}(f(z),k_z) = e^{ik_z z_0} \frac{\int_0^{h_V} f(z)\, e^{ik_z z} dz}{\int_0^{h_V} f(z)\ dz}$$



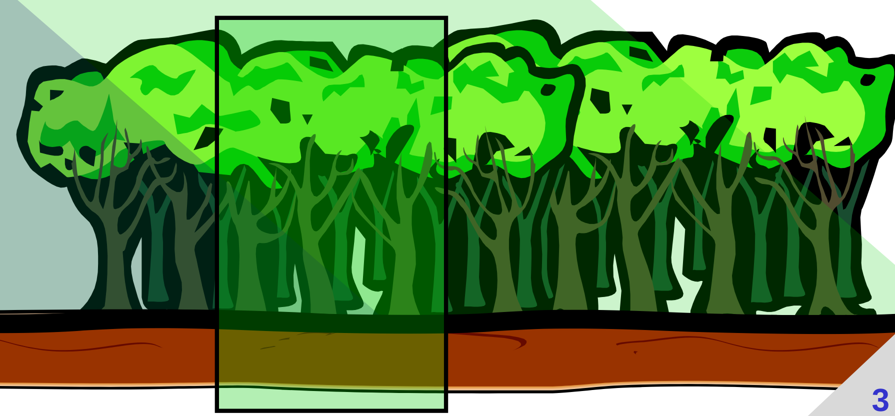


$f(z)$ … vertical reflectivity function

Vertical Wavenumber: $\kappa_z = \dfrac{\kappa \Delta\theta}{\sin(\theta_0)}$

$$\tilde{\gamma} = \tilde{\gamma}_{Temporal}\ \gamma_{SNR}\ \tilde{\gamma}_{Vol}$$

- $\tilde{\gamma}_{Temporal}$ … temporal decorrelation
- γ_{SNR} … additive noise decorrelation
- $\tilde{\gamma}_{Volume}$ … geometric decorrelation



SAR Interferometry for Volume Structure

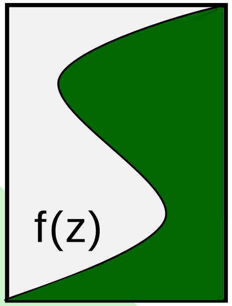
S_1 S_2

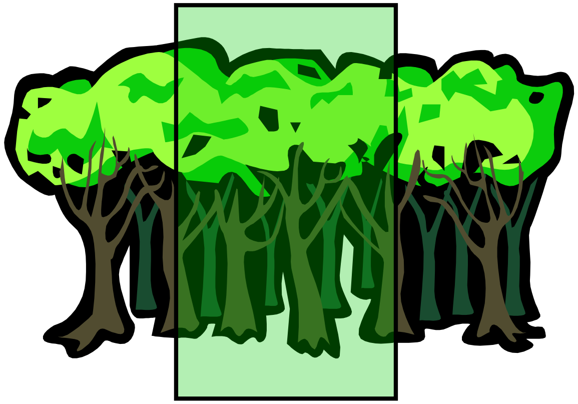
Interferometric Coherence

$$\tilde{\gamma}(S_1\, S_2) = \frac{< S_1\, S_2^* >}{\sqrt{< S_1\, S_1^* > < S_2\, S_2^* >}}$$

Volume Coherence

$$\tilde{\gamma}_{Vol}(f(z),k_z) = e^{ik_z z_0} \frac{\int_0^{h_V} f(z)\, e^{ik_z z} dz}{\int_0^{h_V} f(z)\; dz}$$





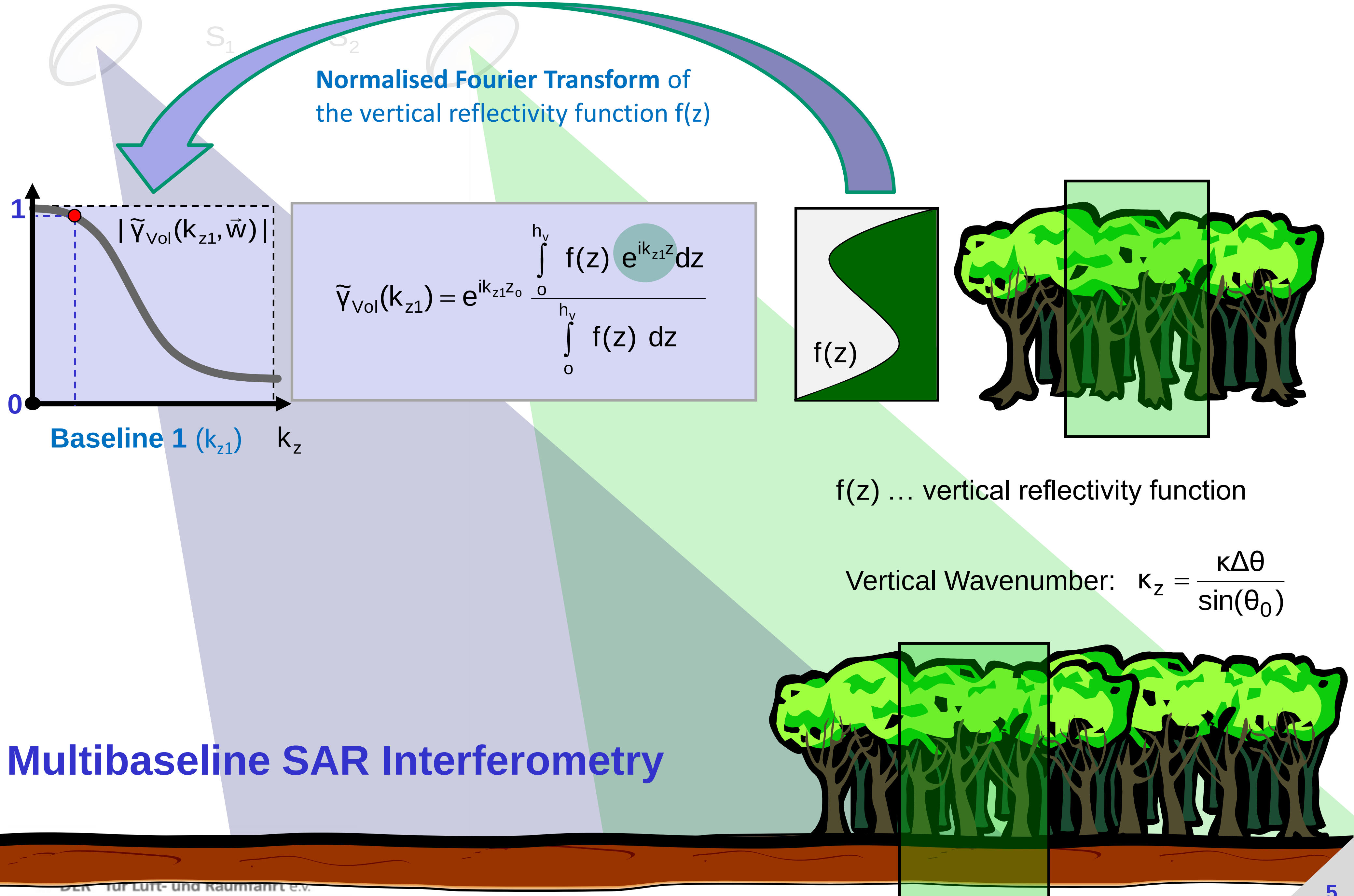
$f(z)$ … vertical reflectivity function

Vertical Wavenumber: $\kappa_z = \dfrac{\kappa \Delta\theta}{\sin(\theta_0)}$

$$\tilde{\gamma} = \tilde{\gamma}_{Temporal}\; \gamma_{SNR}\; \tilde{\gamma}_{Vol}$$

- $\tilde{\gamma}_{Temporal}$ … temporal decorrelation
- γ_{SNR} … additive noise decorrelation
- $\tilde{\gamma}_{Volume}$ … geometric decorrelation

SAR interferometry allows to reconstruct the vertical reflectivity function $f(z)$ of a volume scatterer by means of interferometric (volume) coherence measurements at different vertical wavenumbers κ_Z, i.e. at different spatial baselines.



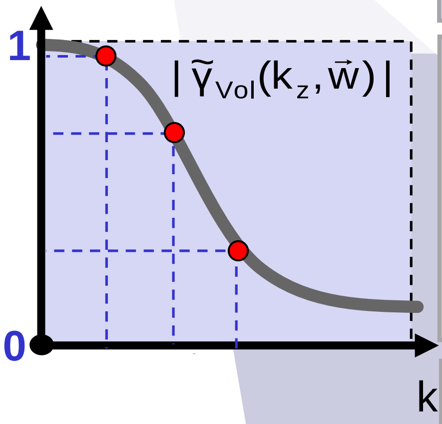
$$\tilde{\gamma}_{Vol}(k_{z1}) = e^{ik_{z1}z_0}\frac{\int_0^{h_v} f(z)\, e^{ik_{z1}z}dz}{\int_0^{h_v} f(z)\ dz}$$

f(z) … vertical reflectivity function

Vertical Wavenumber: $\kappa_z = \dfrac{\kappa\Delta\theta}{\sin(\theta_0)}$

Multibaseline SAR Interferometry

Baseline 3 (k_{z3})

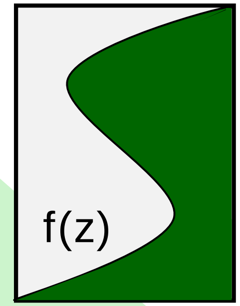


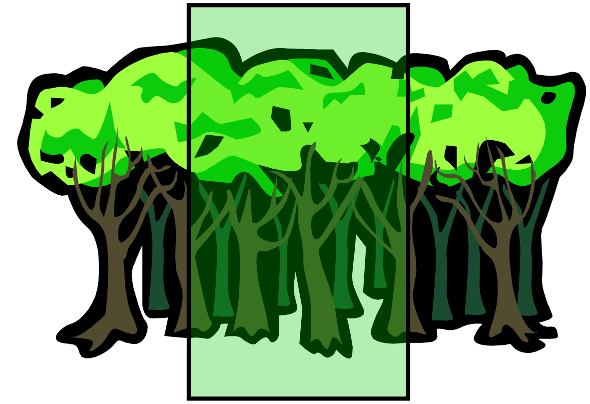
Baseline 2 (k_{z2})

$$\tilde{\gamma}_{Vol}(k_{z3}) = e^{ik_z z_0} \frac{\int_0^{h_V} f(z)\, e^{ik_{z3} z} dz}{\int_0^{h_V} f(z)\, dz}$$

$$\tilde{\gamma}_{Vol}(k_{z1}) = e^{ik_{z1} z_0} \frac{\int_0^{h_V} f(z)\, e^{ik_{z1} z} dz}{\int_0^{h_V} f(z)\, dz}$$

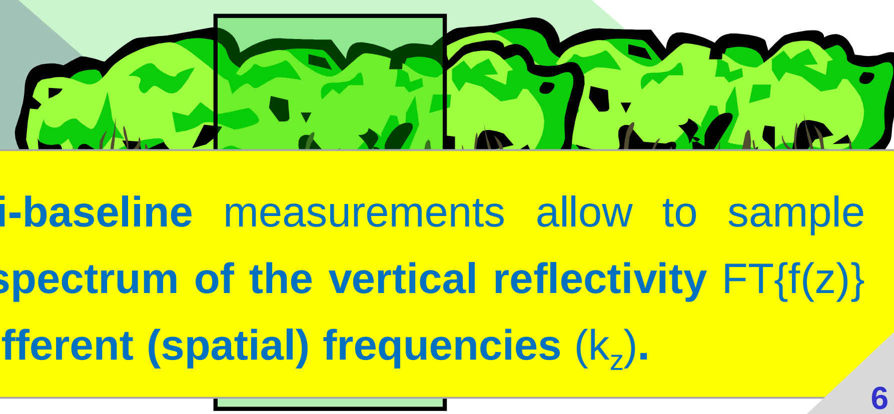
$$\tilde{\gamma}_{Vol}(k_{z2}) = e^{ik_z z_0} \frac{\int_0^{h_V} f(z)\, e^{ik_{z2} z} dz}{\int_0^{h_V} f(z)\, dz}$$





f(z) … vertical reflectivity function

Vertical Wavenumber: $\kappa_z = \frac{\kappa \Delta\theta}{\sin(\theta_0)}$

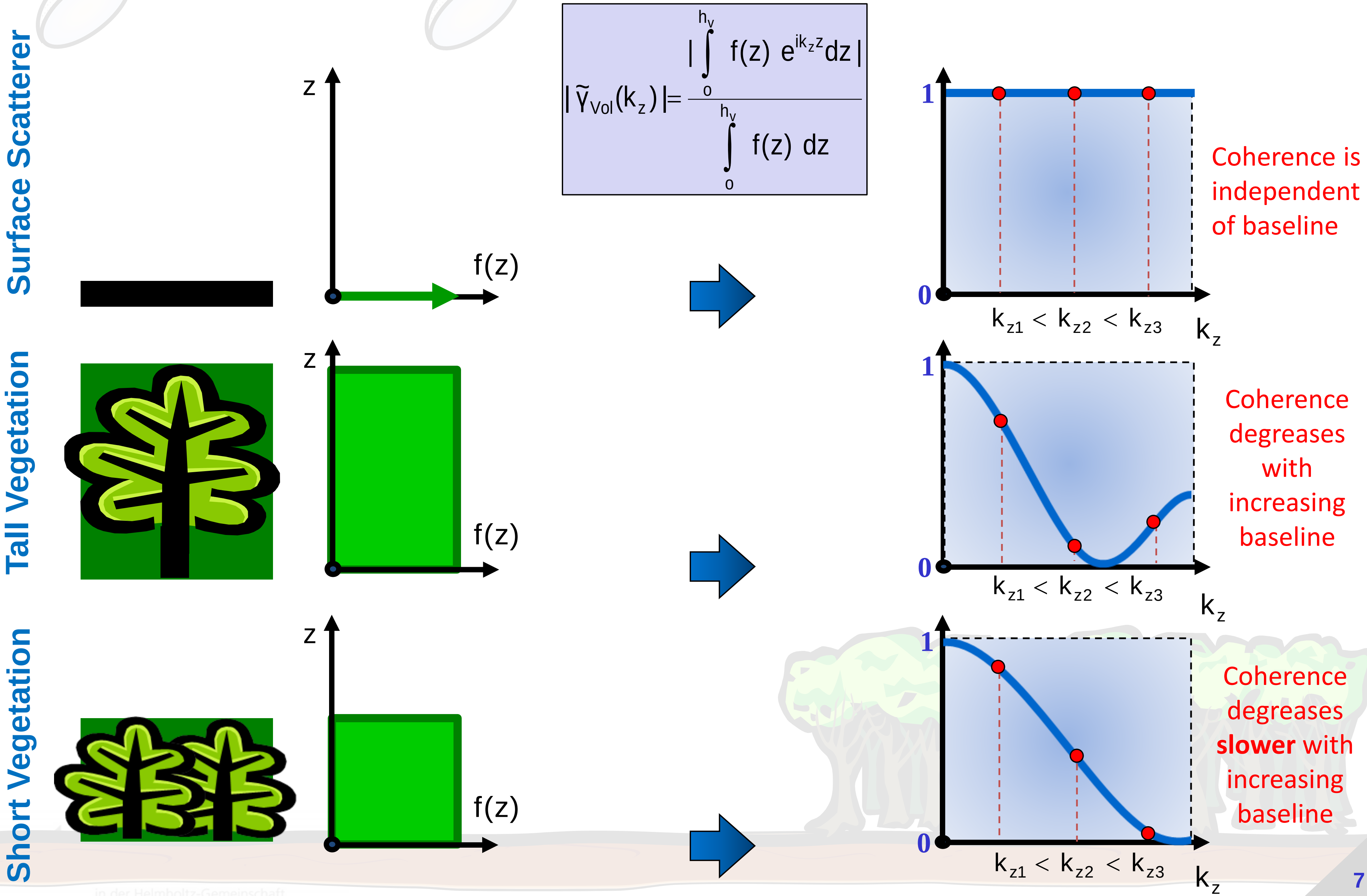


Multibaseline SAR Interferometry

Multi-baseline measurements allow to sample **the spectrum of the vertical reflectivity** FT{f(z)} **@ different (spatial) frequencies** (k_z).

Vertical Reflectivity Function f(z)

InSAR Volume Coherence $|\tilde{\gamma}_{Vol}(k_z)|$



Airborne (E-SAR) Single-Pass Interferometry: X-Band

08biosar0405x1_t01 XTI half baseline

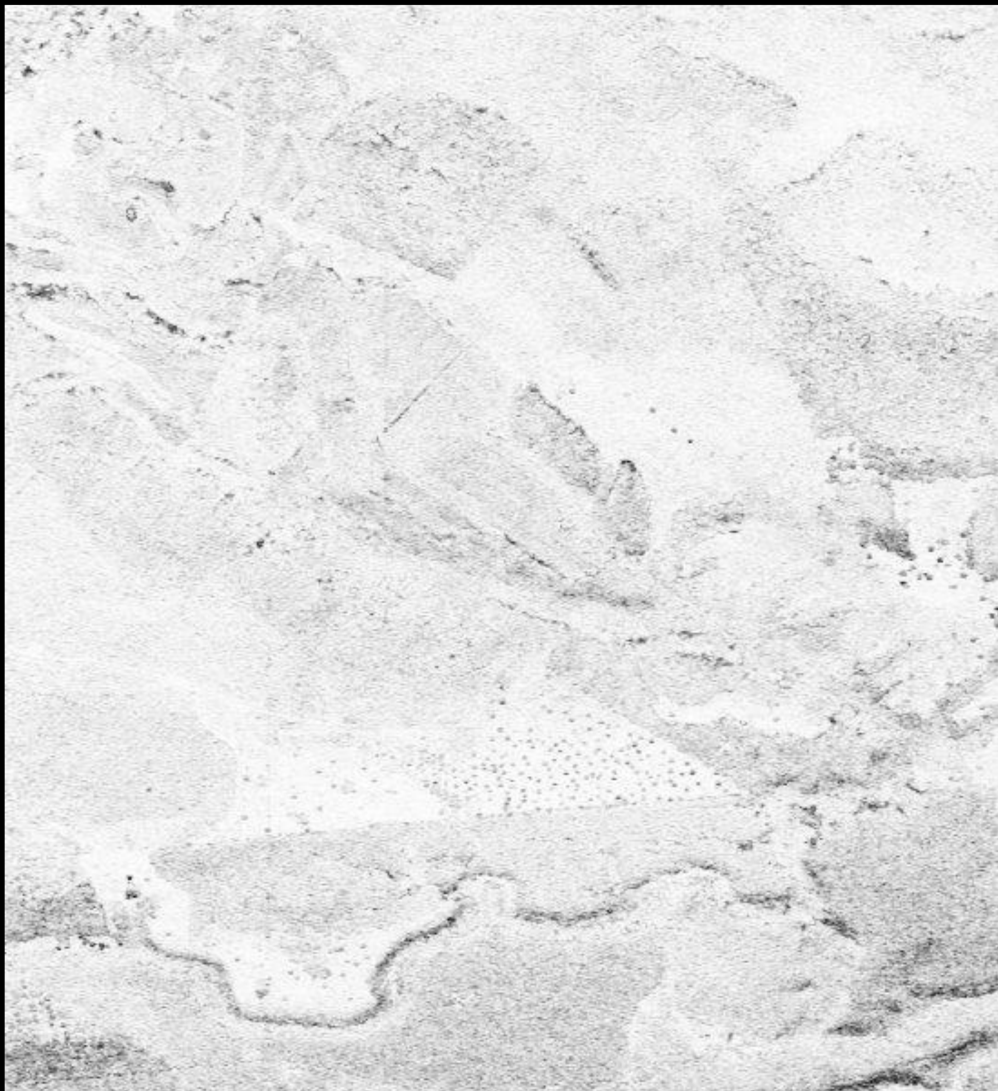
08biosar0407x1_t01 XTI full baseline

Airborne (E-SAR) Single-Pass Interferometry: X-Band

08biosar0405x1_t01 XTI half baseline, k_z = 0.6 rad/m

08biosar0407x1_t01 XTI full baseline, k_z = 1.2 rad/m

Airborne (E-SAR) Single-Pass Interferometry: X-Band



08biosar0405x1_t01 XTI half baseline, k_z = 0.6 rad/m

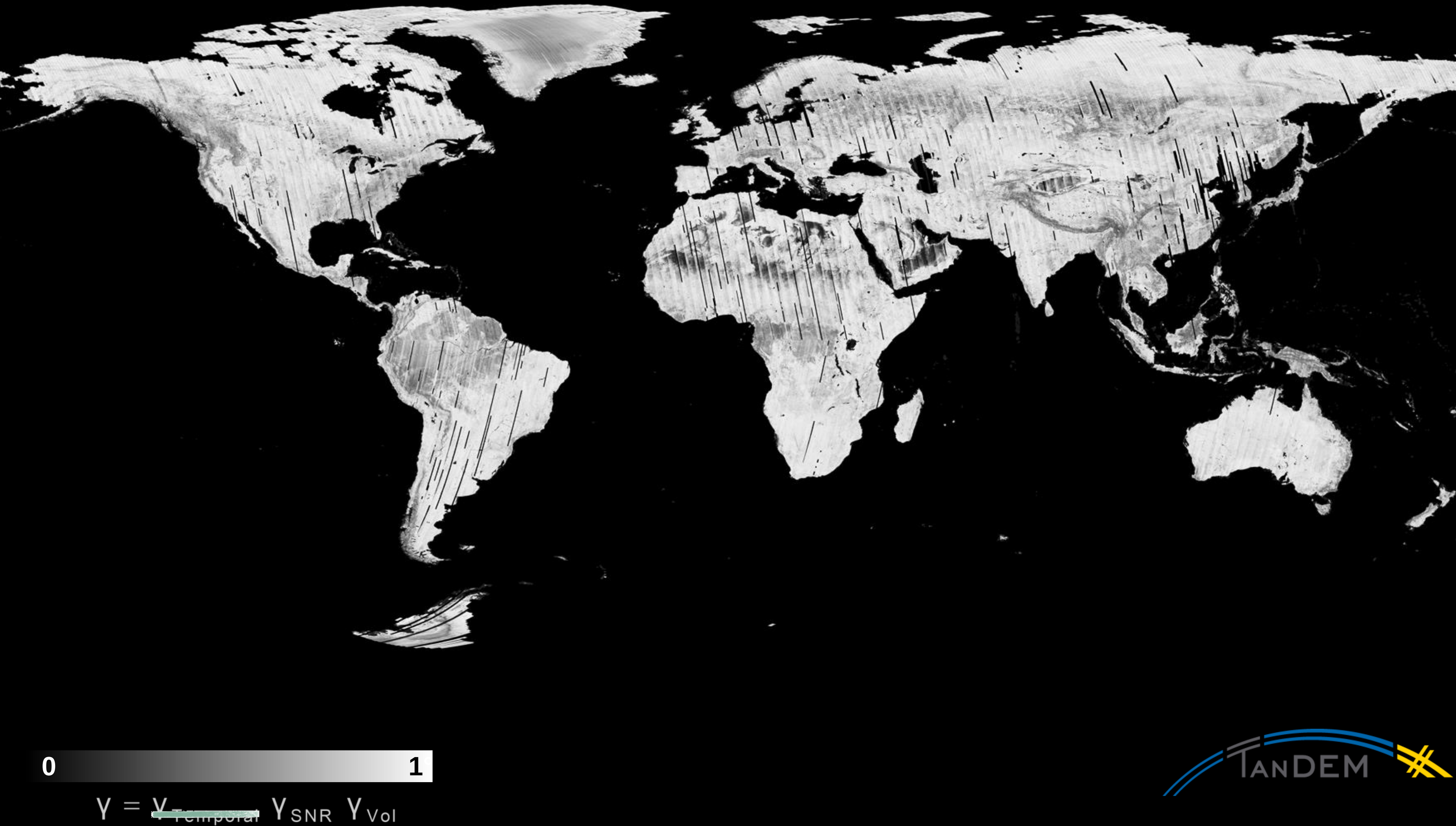
08biosar0407x1_t01 XTI full baseline, k_z = 1.2 rad/m

$$\gamma = \cancel{\gamma_{Temporal}}\ \cancel{\gamma_{SNR}}\ \gamma_{Vol}$$

Airborne (E-SAR) Single-Pass Interferometry: X-Band

08biosar0405x1_t01 XTI half baseline

08biosar0407x1_t01 XTI full baseline



By DLR-HR-STL

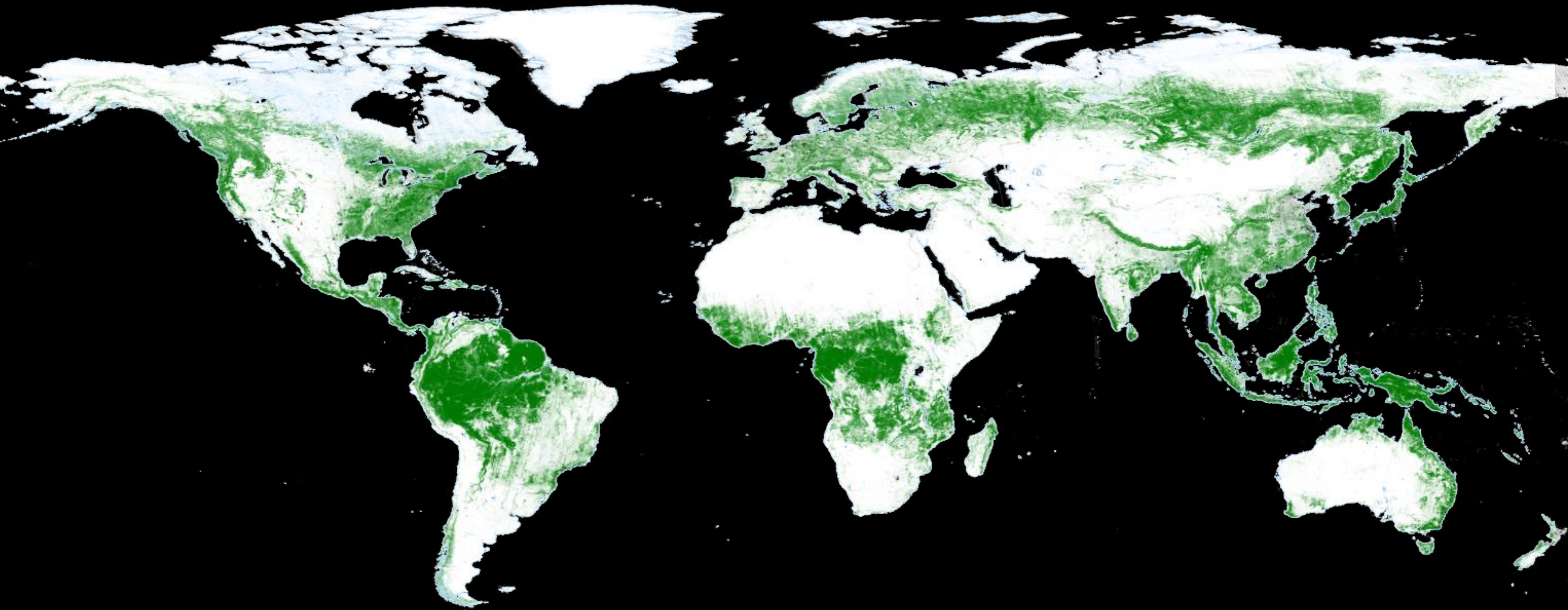
500x500 m^2 resolution

0 1

$\gamma = \cancel{\gamma_{Temporal}} \; \gamma_{SNR} \; \gamma_{Vol}$

By DLR-HR-STL

500x500 m^2 resolution



Forest – Non Forest Map

By DLR-HR-STL

50x50 m^2 resolution

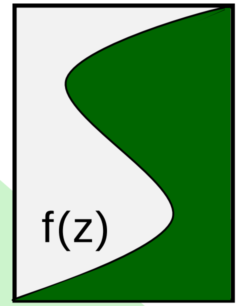
Interferometric Coherence

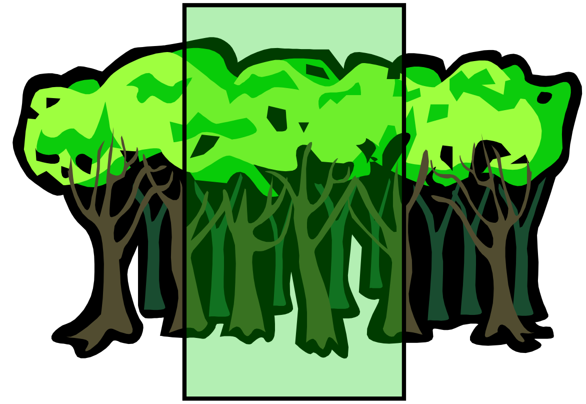
$$\tilde{\gamma}(S_1\ S_2) = \frac{< S_1\ S_2^* >}{\sqrt{< S_1\ S_1^* > < S_2\ S_2^* >}}$$

SAR Interferometry for Volume Structure

Volume Coherence

$$\tilde{\gamma}_{Vol}(f(z),k_z) = e^{ik_z z_0} \frac{\int_0^{h_V} f(z)\ e^{ik_z z} dz}{\int_0^{h_V} f(z)\ dz}$$





$f(z)$ … vertical reflectivity function

Vertical Wavenumber: $\kappa_z = \dfrac{\kappa \Delta \theta}{\sin(\theta_0)}$

$$\tilde{\gamma} = \tilde{\gamma}_{Temporal}\ \gamma_{SNR}\ \tilde{\gamma}_{Vol}$$

- $\tilde{\gamma}_{Temporal}$ … temporal decorrelation
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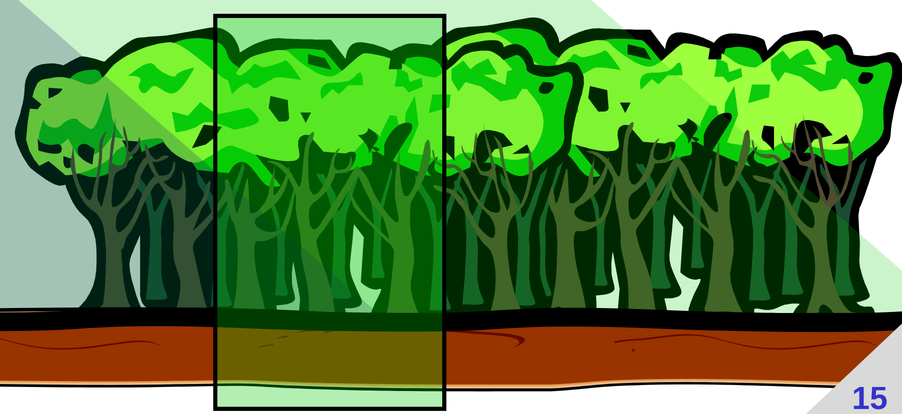
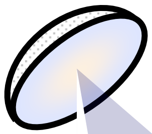
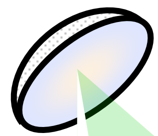
- $\tilde{\gamma}_{Volume}$ … geometric decorrelation







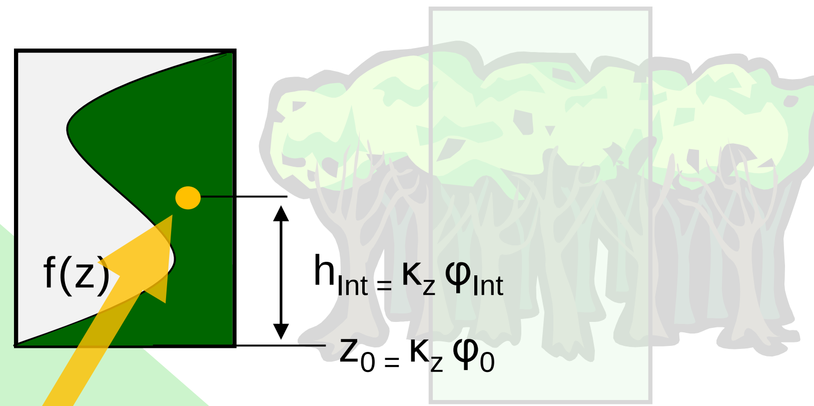
Interferometric Coherence

$$\tilde{\gamma}(S_1\ S_2) = \frac{< S_1\ S_2^* >}{\sqrt{< S_1\ S_1^* > < S_2\ S_2^* >}}$$

SAR Interferometry for Volume Structure

Volume Coherence

$$\tilde{\gamma}_{Vol}(f(z),k_z) = e^{ik_z z_0} \frac{\int_0^{h_V} f(z)\, e^{ik_z z} dz}{\int_0^{h_V} f(z)\, dz}$$



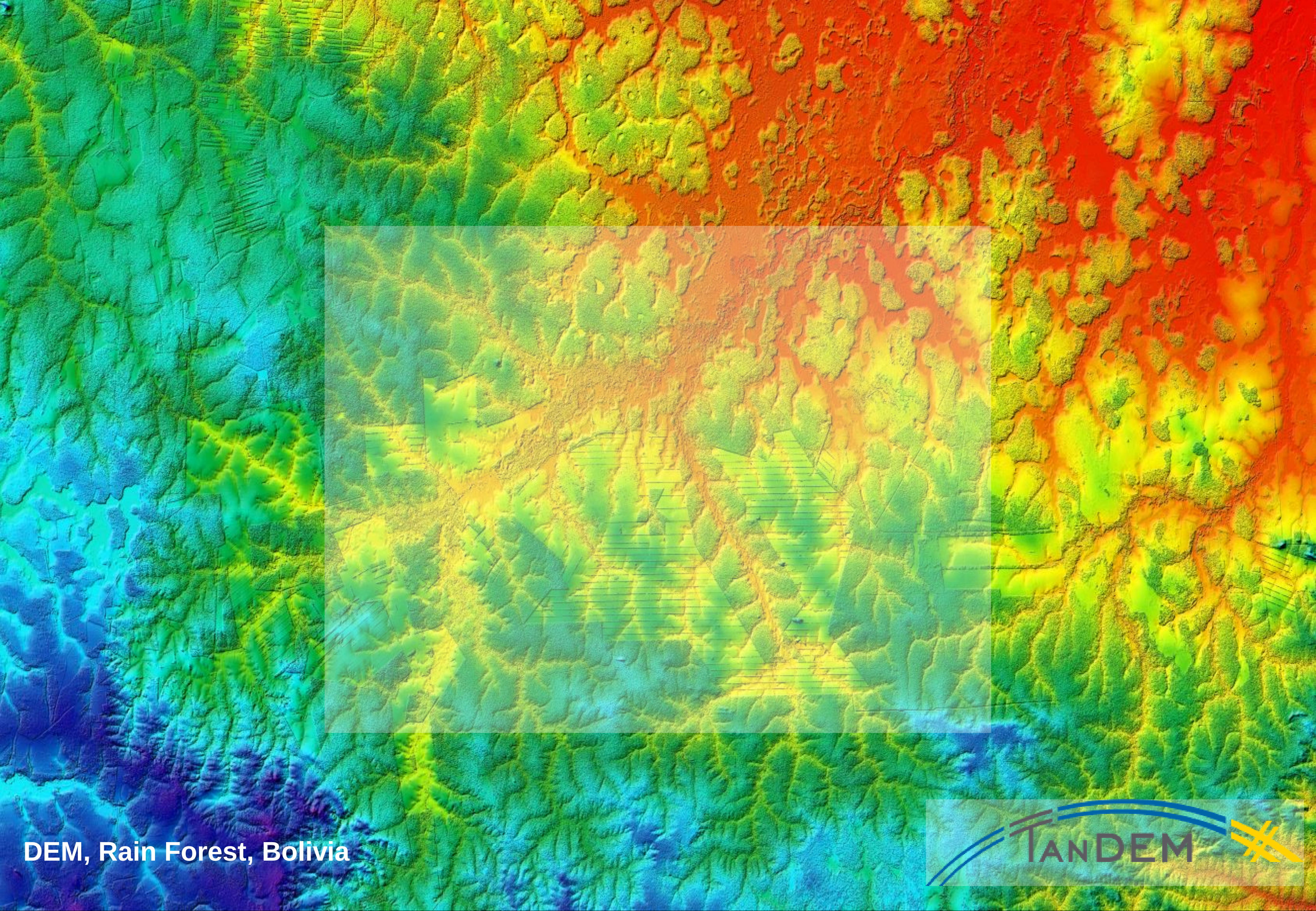
$f(z)$ … vertical reflectivity function

Vertical Wavenumber: $\kappa_z = \dfrac{\kappa \Delta\theta}{\sin(\theta_0)}$

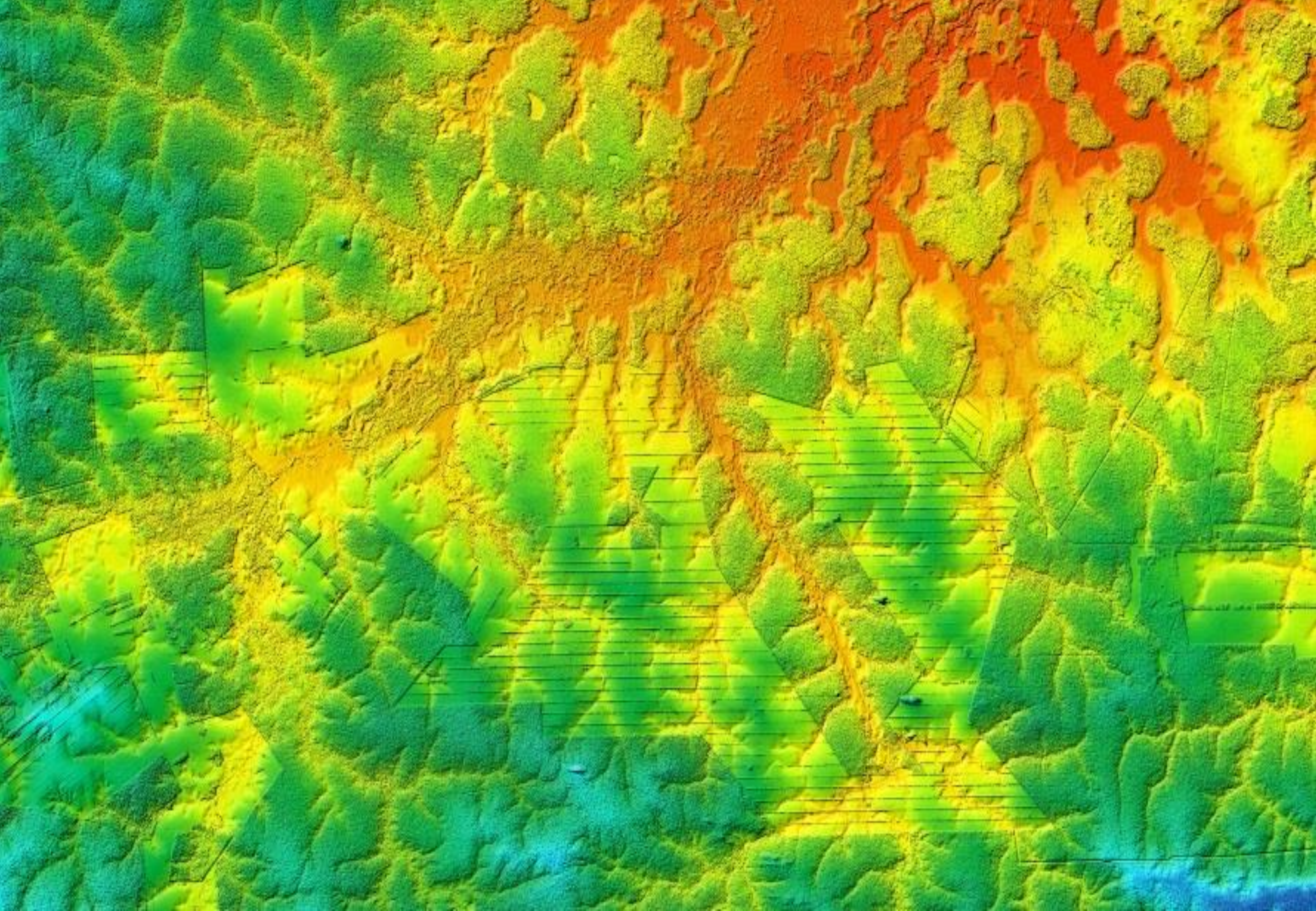
$$\tilde{\gamma} = \tilde{\gamma}_{Temporal}\ \gamma_{SNR}\ \tilde{\gamma}_{Vol}$$

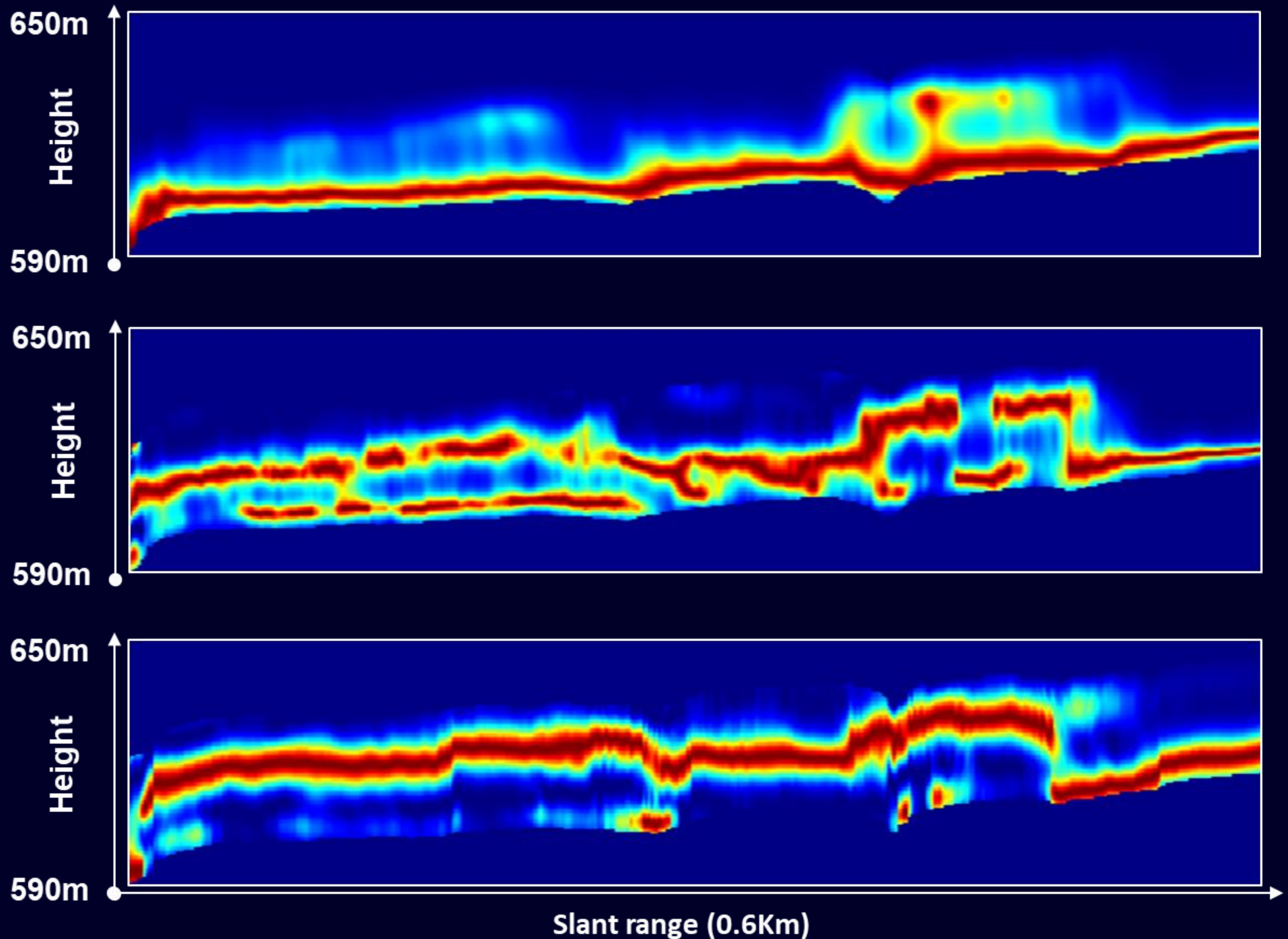
- $\tilde{\gamma}_{Temporal}$ … temporal decorrelation
- γ_{SNR} … additive noise decorrelation
- $\tilde{\gamma}_{Volume}$ … geometric decorrelation

The **phase (center)** of is γ_{Vol} associated to the **center of mass of f(z)**: The phase center height $h_{Int} = \kappa_z\, \varphi_{Int}$ corresponds to the height of the center of mass of f(z) with respect to z_0 !!!



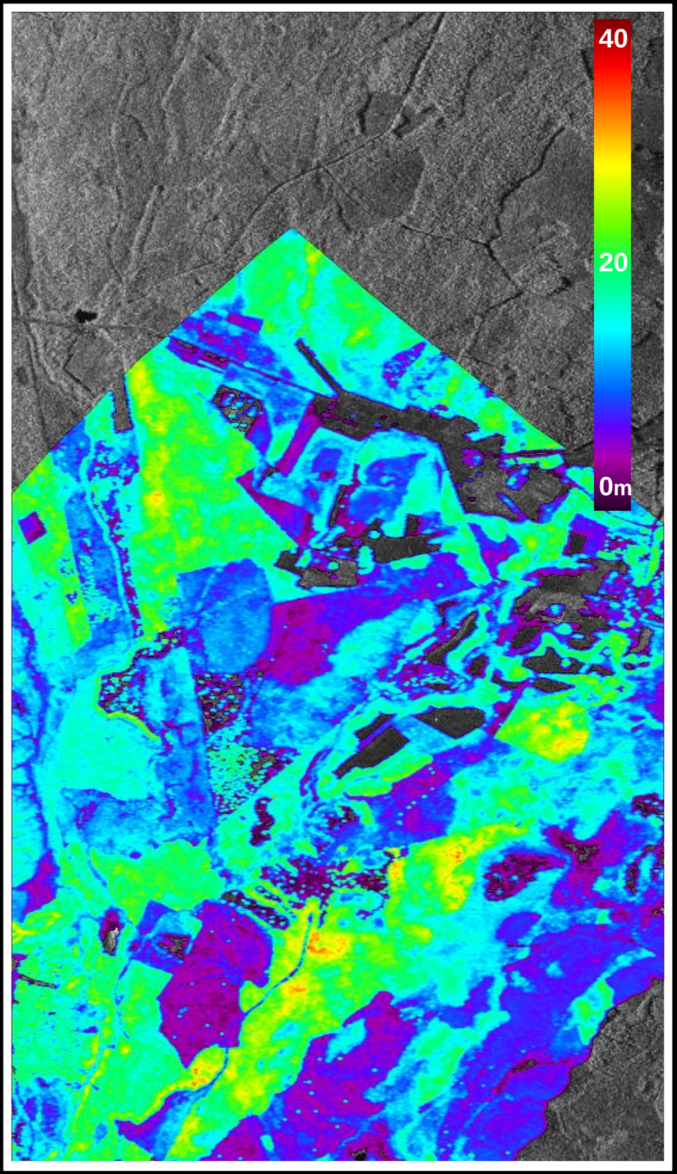
DEM, Rain Forest, Bolivia



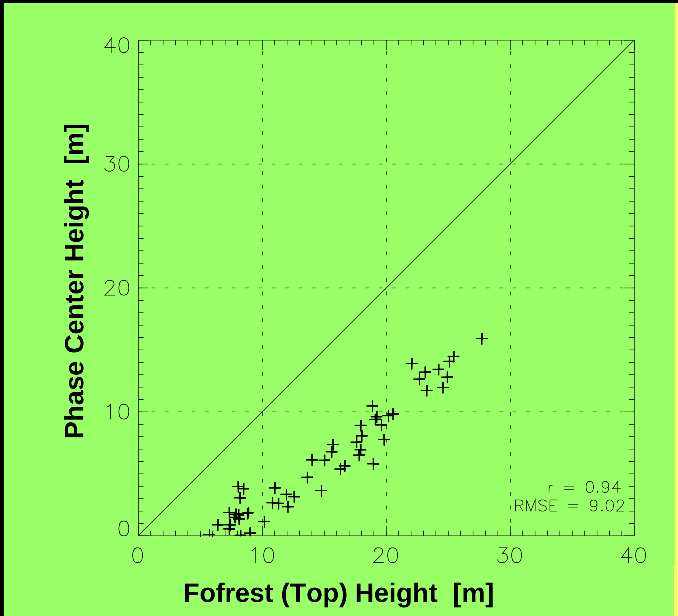


Traunstein forest (Germany) - Capon - HH

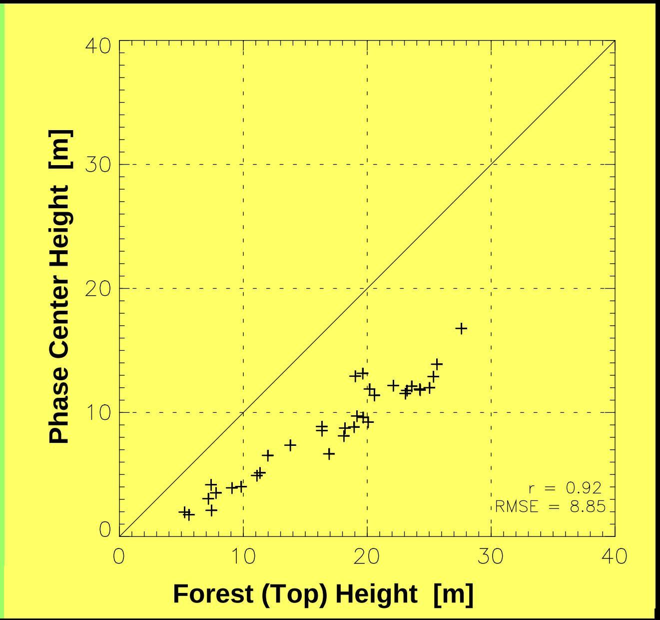
Test Site: Krycklan, Sweden



Half Baseline



Full Baseline



DLR

Forest Height inversion from InSAR Data

$$\tilde{\gamma}(s_1\, s_2) = \frac{\langle s_1\, s_2^* \rangle}{\sqrt{\langle s_1\, s_1^* \rangle \langle s_2\, s_2^* \rangle}}$$

Interferometric Coherence (complex)

$$\tilde{\gamma}(s_1\, s_2) = \gamma_{Sys}\, \gamma_{Tmp}\, \tilde{\gamma}_{Vol}$$

$\tilde{\gamma}_{Vol}$ … volume decorrelation $\tilde{\gamma}_{Tmp}$ … temporal decorrelation

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = e^{ik_z z_0} \frac{\int_0^{h_v} f(z,\vec{w})\, e^{i\kappa_z z} dz}{\int_0^{h_v} f(z,\vec{w})\, dz}$$

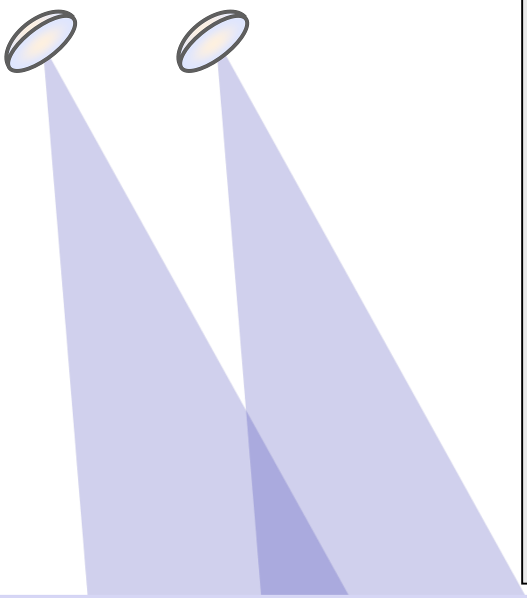
$f(z,\vec{w})$ … vertical reflectivity function κ_z … vertical wavenumber

Interferometric (volume) decorrelation is sensitive to the „visible“ (forest) **height** and to the **vertical reflectivity function** $f(z,\vec{w})$ within the interferometric resolution cell.

Forest Height inversion challenges:

- The parameterisation / description of $f(z,\vec{w})$
- The presence of $\tilde{\gamma}_{Tmp}$

2 Layer Inversion Model: (Random) Volume over Ground RVoG



Volume Layer Ground Layer

$$f(z,\vec{w}) = \underbrace{m_V f_V(z)}_{\text{Volume Layer}} + \underbrace{m_G(\vec{w})\delta(z-z_0)}_{\text{Ground Layer}}$$

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = \exp(i\varphi_0)\frac{\tilde{\gamma}_V(\kappa_z)+m(\vec{w})}{1+m(\vec{w})}$$

Volume Layer Coherence

$$\tilde{\gamma}_V = \frac{I}{I_0} \quad \begin{cases} I = \int_0^{h_V} \exp(i\kappa_z z') f_V(z')dz' \\ I_0 = \int_0^{h_V} f_V(z')dz' \end{cases}$$

$$m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0} \qquad \kappa_z = \frac{\kappa\Delta\theta}{\sin(\theta_0)}$$

$f_V(z)$ … volume reflectivity function

$\varphi_0 = k_z z_0$ … underlying topography

Single Baseline Observations

single- / dual- / quad-pol

$\tilde{\gamma}_{Vol}(\vec{w}_1,\kappa_z)$ $\tilde{\gamma}_{Vol}(\vec{w}_2,\kappa_z)$ $\tilde{\gamma}_{Vol}(\vec{w}_3,\kappa_z)$

1, 2, or 3 complex coherences

Total Coherence

$$\tilde{\gamma}(\vec{w},\kappa_z) = \gamma_{Temp}(\kappa_z)\,\tilde{\gamma}_{Vol}(\vec{w},\kappa_z)$$

For a Single Baseline

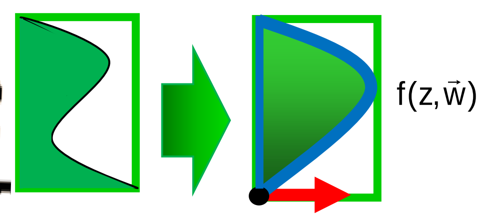
3+N unknown parameters

Volume Height	h_V
Topography	φ_0
G/V Ratio	$m(\vec{w}) = f(pol)$
$f_V(z)$	… N parameters

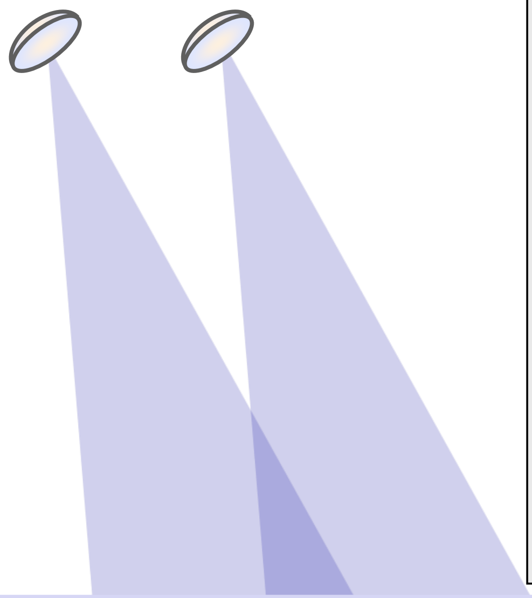
plus one more

Temporal Deco	γ_{Temp}

for repeat-pass implementations



2 Layer Inversion Model with exponential volume reflectivity



Volume Layer   Ground Layer

$$f(z,\vec{w}) = \boxed{m_V f_V(z)} + \boxed{m_G(\vec{w})\delta(z-z_0)}$$

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = \exp(i\varphi_0)\frac{\tilde{\gamma}_V(\kappa_z) + m(\vec{w})}{1+m(\vec{w})}$$

Volume Layer Coherence

$$\tilde{\gamma}_V = \frac{I}{I_0} \begin{cases} I = \int_0^{h_V} \exp(i\kappa_z z') e^{\left(\frac{2\ \sigma\ z'}{\cos\theta_0}\right)} dz' \\ I_0 = \int_0^{h_V} e^{\left(\frac{2\ \sigma\ z'}{\cos\theta_0}\right)} dz' \end{cases}$$

$$m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0} \qquad \kappa_z = \frac{\kappa\Delta\theta}{\sin(\theta_0)}$$

$$f_V(z) = \exp\left(\frac{2\ \sigma\ z}{\cos\theta_0}\right)$$

exponential volume reflectivity

Single Baseline Observations

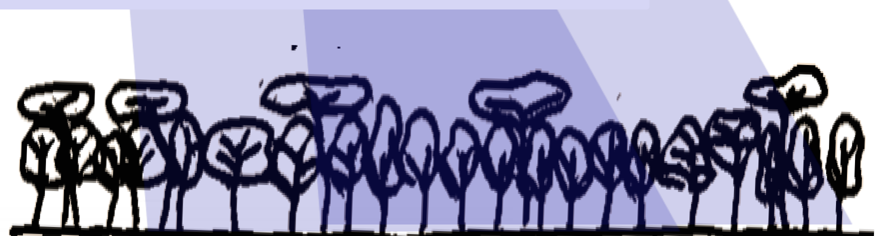
single- / dual- / quad-pol

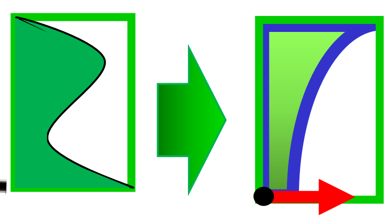
$\tilde{\gamma}_{Vol}(\vec{w}_1,\kappa_z) \quad \tilde{\gamma}_{Vol}(\vec{w}_2,\kappa_z) \quad \tilde{\gamma}_{Vol}(\vec{w}_3,\kappa_z)$

1, 2, or 3 complex coherences

Total Coherence

$$\tilde{\gamma}(\vec{w},\kappa_z) = \gamma_{Temp}(\kappa_z)\ \tilde{\gamma}_{Vol}(\vec{w},\kappa_z)$$





For a Single Baseline

4 unknown parameters

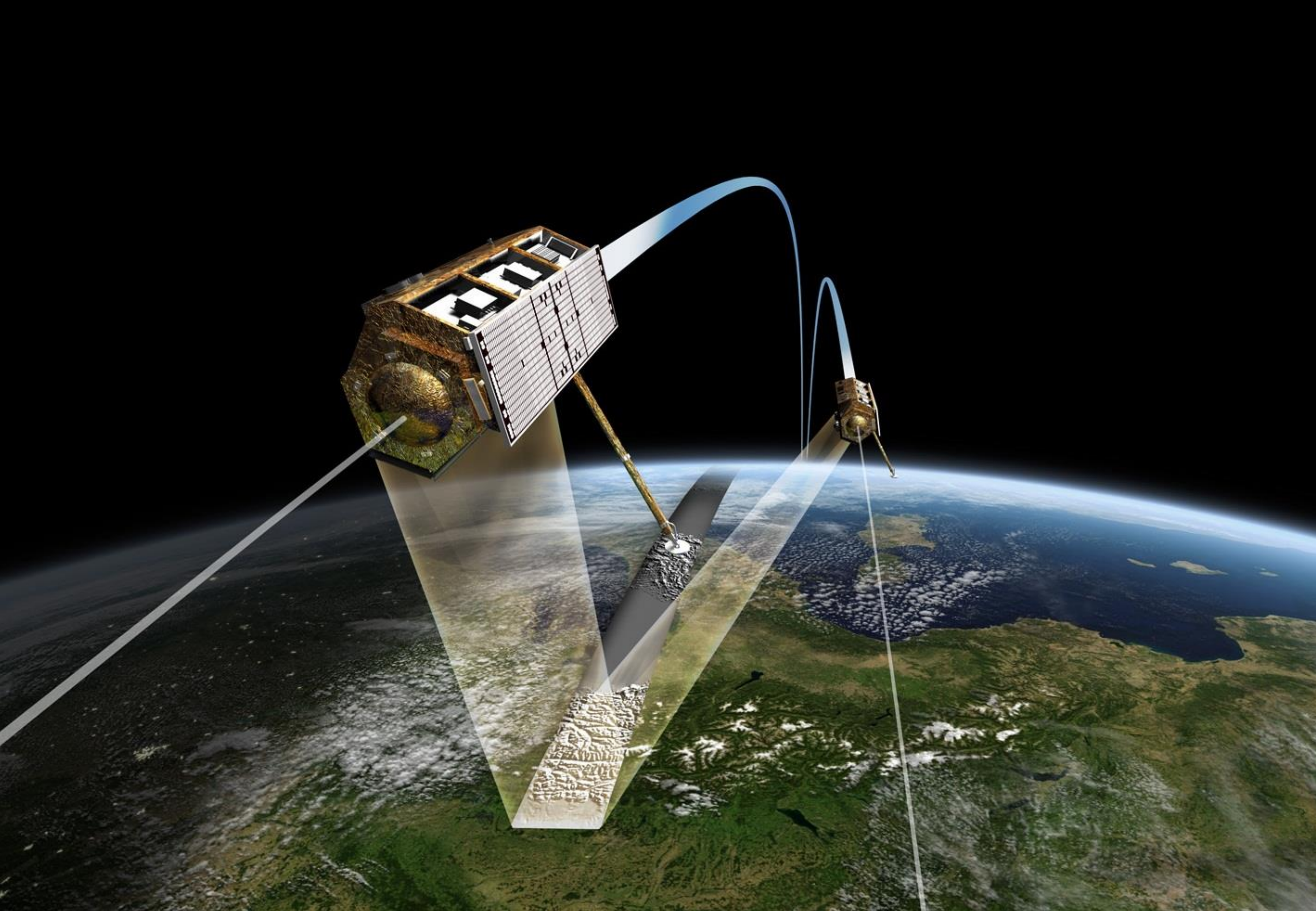
- Volume Height h_V
- Topography φ_0
- G/V Ratio $m(\vec{w}) = f(pol)$
- $f_V(z)$ … 1 parameter (σ)

plus one more

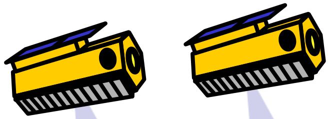
- Temporal Deco γ_{Temp}

for repeat-pass implementations

Deutsches Zentrum für Luft- und Raumfahrt e.V. in der Helmholtz-Gemeinschaft
DLR



TanDEM-X: 2 Layer model with exp. volume reflectivity



Volume Layer   Ground Layer

$$f(z,\vec{w}) = m_V f_V(z) + m_G(\vec{w})\delta(z - z_0)$$

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = \exp(i\varphi_0)\frac{\tilde{\gamma}_V(\kappa_z) + m(\vec{w})}{1 + m(\vec{w})}$$

Volume Layer Coherence

$$\tilde{\gamma}_V = \frac{I}{I_0} \quad \begin{cases} I = \int_0^{h_V} \exp(i\kappa_z z') e^{\left(\frac{2\ \sigma\ z'}{\cos\theta_0}\right)} dz' \\ I_0 = \int_0^{h_V} e^{\left(\frac{2\ \sigma\ z'}{\cos\theta_0}\right)} dz' \end{cases}$$

$$m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0} \qquad \kappa_z = \frac{\kappa\Delta\theta}{\sin(\theta_0)}$$

$$f_V(z) = \exp\left(\frac{2\ \sigma\ z}{\cos\theta_0}\right)$$

exponential volume reflectivity

Total Coherence

$$\tilde{\gamma}(\vec{w},\kappa_z) = \cancel{\gamma_{Temp}(\kappa_z)}\ \tilde{\gamma}_{Vol}(\vec{w},\kappa_z)$$

Single Baseline Observation(s)

single-pol

$\tilde{\gamma}_{Vol}(\vec{w}_1,\kappa_z)$

1 complex coherence

For a Single Baseline

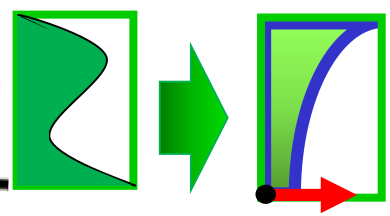
4 unknown parameters

Volume Height h_V

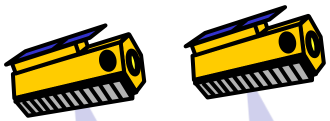
Topography φ_0

G/V Ratio $m(\vec{w})$ = f(pol)

$f_V(z)$ … 1 parameter (σ)



TanDEM-X: 2 Layer model with exp. volume reflectivity + no ground



Invertible with single-pol data only if the ground topography φ_0 is known !!!

Volume Layer   Ground Layer

$$f(z,\vec{w}) = m_V f_V(z) + \cancel{m_G(\vec{w})\delta(z-z_0)}$$

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = \exp(i\varphi_0)\frac{\tilde{\gamma}_V(\kappa_z) + \cancel{m(\vec{w})}}{\cancel{1+m(\vec{w})}}$$

Volume Layer Coherence

$$\tilde{\gamma}_V = \frac{I}{I_0} \begin{cases} I = \int_0^{h_V} \exp(i\kappa_z z') e^{\left(\frac{2\sigma z'}{\cos\theta_0}\right)} dz' \\ I_0 = \int_0^{h_V} e^{\left(\frac{2\sigma z'}{\cos\theta_0}\right)} dz' \end{cases}$$

$$\cancel{m(\vec{w}) = \frac{m_G(\vec{w})}{m_V(\vec{w}) I_0}} \qquad \kappa_z = \frac{\kappa\Delta\theta}{\sin(\theta_0)}$$

$$f_V(z) = \exp\left(\frac{2\sigma z}{\cos\theta_0}\right)$$

exponential volume reflectivity

Total Coherence

$$\tilde{\gamma}(\vec{w},\kappa_z) = \cancel{\gamma_{Temp}(\kappa_z)}\ \tilde{\gamma}_{Vol}(\vec{w},\kappa_z)$$

Single Baseline Observation(s)

single-pol

$\tilde{\gamma}_{Vol}(\vec{w}_1,\kappa_z)$

1 complex coherence

For a Single Baseline

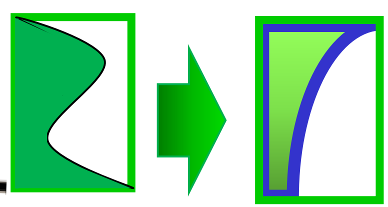
3 unknown parameters

Volume Height h_V

Topography φ_0

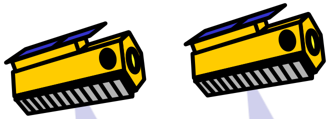
~~G/V Ratio~~ $\cancel{m(\vec{w}) \quad f(pol)}$

$f_V(z)$ … 1 parameter (σ)



$f_V(z)$

TanDEM-X: 2 Layer model with ct. volume reflectivity + no ground



Volume Layer Ground Layer

$$f(z,\vec{w}) = m_V f_V(z) + \cancel{m_G(\vec{w})\delta(z-z_0)}$$

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = \exp(i\varphi_0)\frac{\tilde{\gamma}_V(\kappa_z) + \cancel{m(\vec{w})}}{\cancel{1+m(\vec{w})}}$$

Volume Layer Coherence

$$\tilde{\gamma}_V = \exp\left(i\frac{h_V\kappa_z}{2}\right)\frac{\sin\left(\frac{h_V\kappa_z}{2}\right)}{\left(\frac{h_V\kappa_z}{2}\right)}$$

$$\cancel{m(\vec{w})} = \frac{m_G(\vec{w})}{m_V(\vec{w})I_0} \qquad \kappa_z = \frac{\kappa\Delta\theta}{\sin(\theta_0)}$$

$f_V(z) = ct.$ constant volume reflectivity (box profile)

Single Baseline Observation(s)

single-pol ✓

$\tilde{\gamma}_{Vol}(\vec{w}_1,\kappa_z)$

1 complex coherence

Total Coherence

$$\tilde{\gamma}(\vec{w},\kappa_z) = \cancel{\gamma_{Temp}(\kappa_z)}\,\tilde{\gamma}_{Vol}(\vec{w},\kappa_z)$$

For a Single Baseline

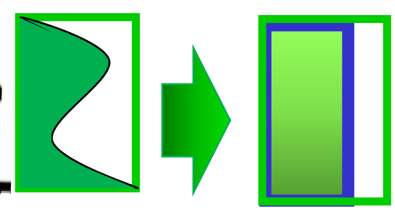
2 unknown parameters

Volume Height h_V

Topography φ_0

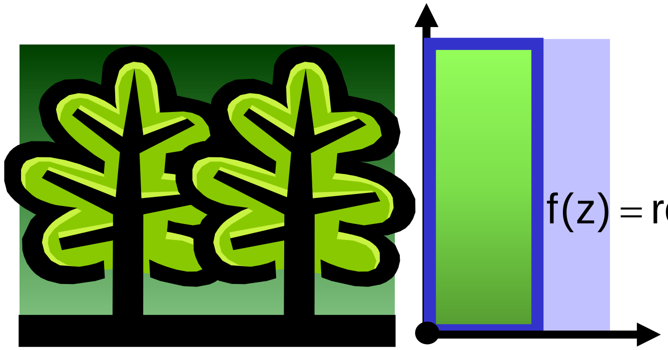
~~G/V Ratio $m(\vec{w})$ f(pol)~~

~~$f_V(z)$ → 1-2 parameter(s)~~



$f_V(z)$

Box-Profile and Sinc-Inversion



$$f(z) = \mathrm{rect}(z) = \begin{cases} \sigma_{V0} & \text{when } 0 \le z \le h_V \\ 0 & \text{when } \quad z > h_V \end{cases}$$

$$\tilde{\gamma}_{Vol}(\vec{w},\kappa_z) = \exp i(\varphi_0 + \frac{h_V\kappa_z}{2}) \frac{\sin\left(\frac{h_V\kappa_z}{2}\right)}{\left(\frac{h_V\kappa_z}{2}\right)}$$

Total Coherence: $\tilde{\gamma}(\vec{w},\kappa_z) = \gamma_{Sys}\,\tilde{\gamma}_{Vol}(\vec{w},\kappa_z)$

System Decorrelation: $\gamma_{Sys} = 0.97$

Volume (Forest) Height: h_V

Topography: ϕ_0

Vertical Wavenumber: $\kappa_z = \frac{\kappa\Delta\theta}{\sin(\theta_0)}$

Inversion: $$|\tilde{\gamma}_{Vol}(\kappa_z)| = \frac{|\tilde{\gamma}(\vec{w},\kappa_z)|}{|\gamma_{Sys}|} = \left|\frac{\sin\left(\frac{h_V\kappa_z}{2}\right)}{\left(\frac{h_V\kappa_z}{2}\right)}\right| \rightarrow h_V \approx \frac{2\pi}{\kappa_z}\left[1 - \frac{2}{\pi}\sin^{-1}\left(\frac{|\tilde{\gamma}(\vec{w},\kappa_z)|^{0.8}}{|\gamma_{Sys}|^{0.8}}\right)\right]$$

The Pol-InSAR Course

Part 5: SAR Interferometry 3

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